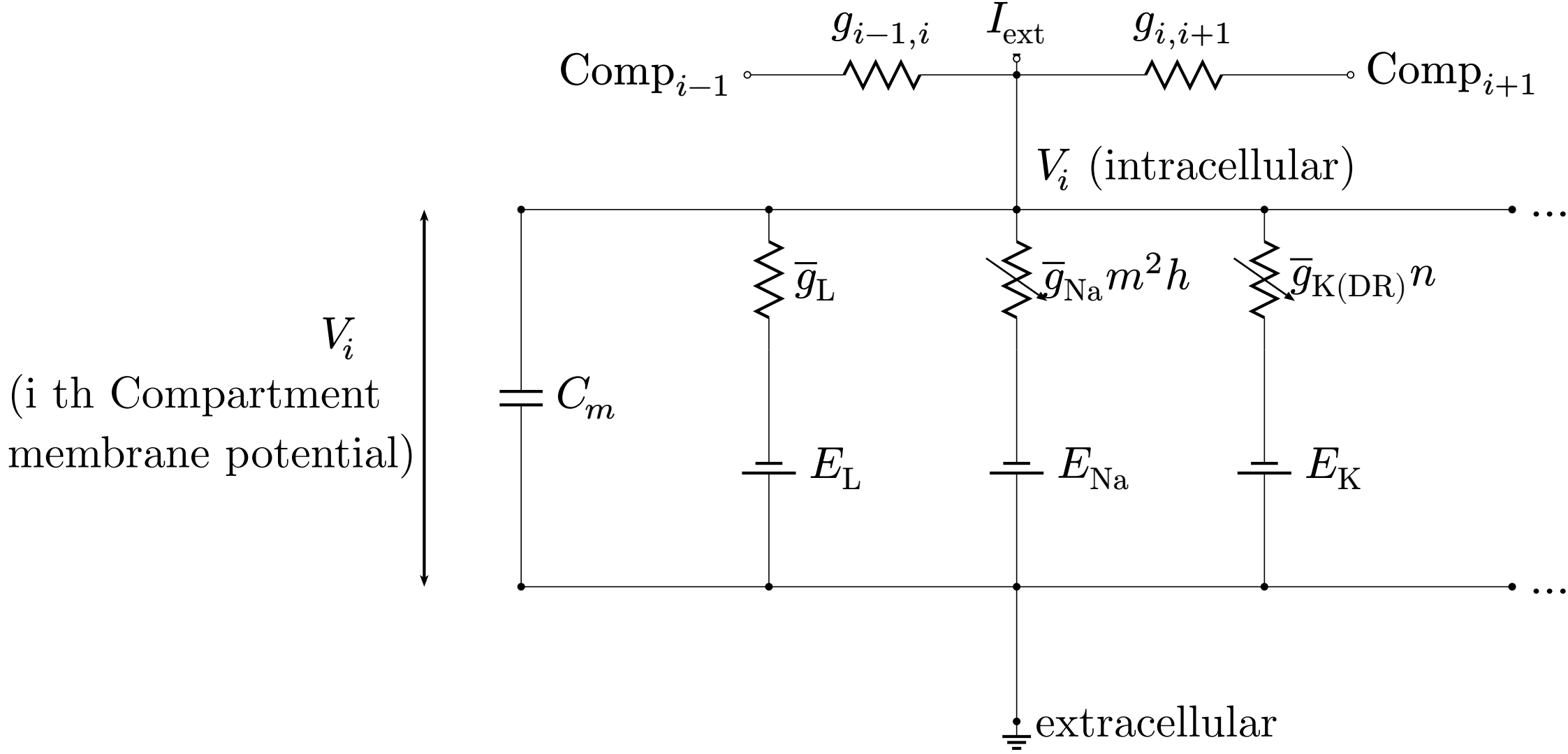
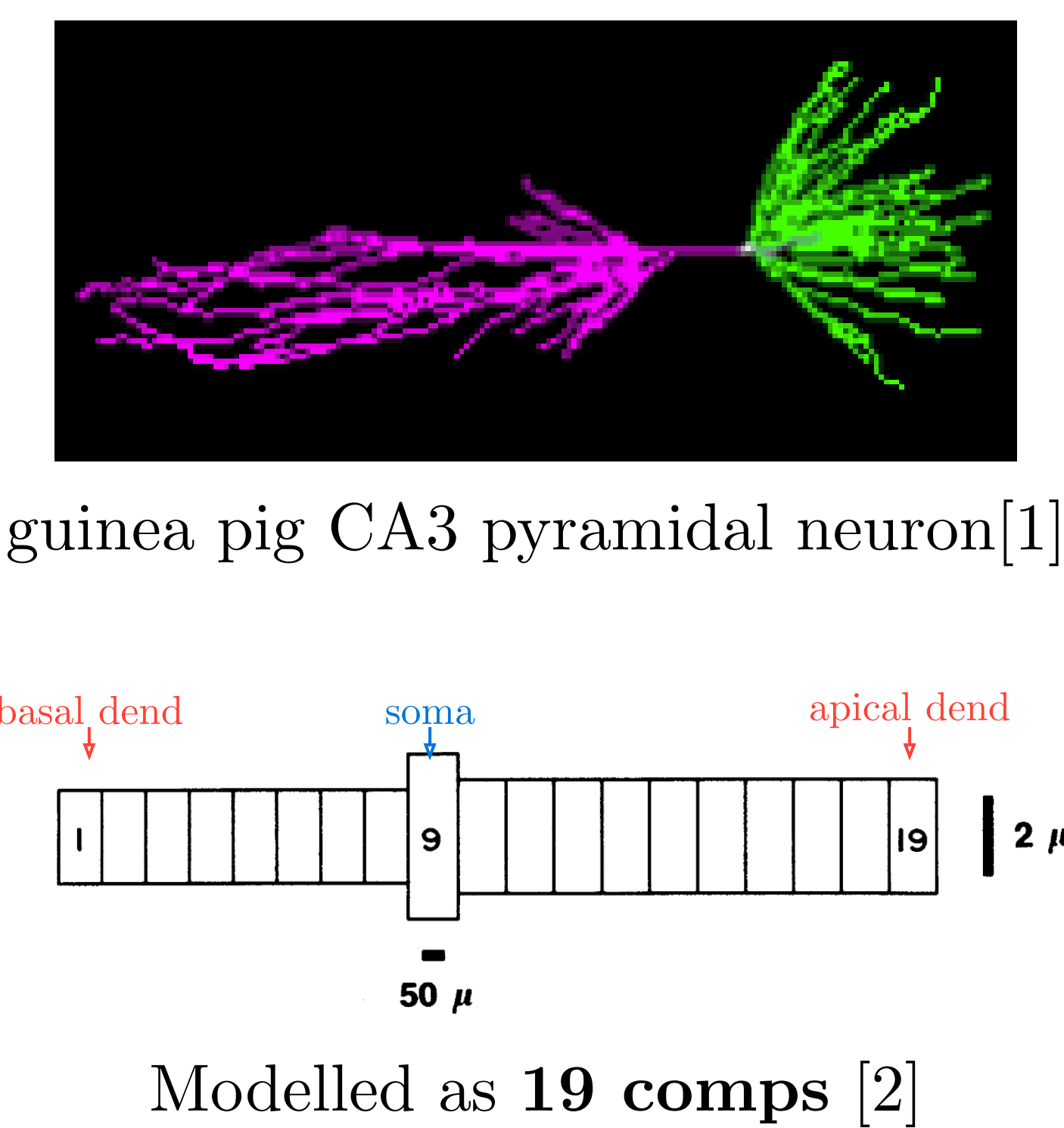


Introduction



Conductances have **gate variables**, ODEs with rate functions α, β :

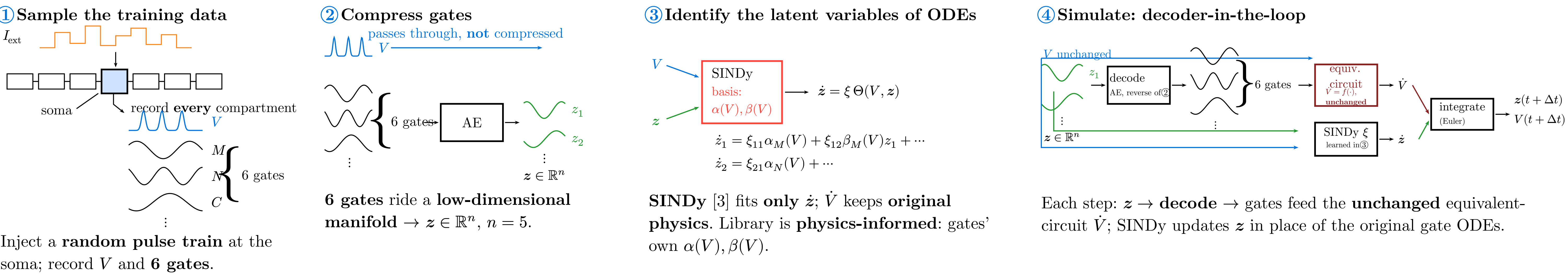
$$I_{Na} = \bar{g}_{Na} m^2 h (V - E_{Na})$$
$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m$$
$$\alpha_m(V) = 0.32(13.1 - u)/(\exp((13.1 - u)/4) - 1)$$
$$\beta_m(V) = 0.28(u - 40.1)/(\exp((u - 40.1)/5) - 1)$$

→ **11 states per compartment**
→ **209** for 19 comps;
At large scale network simulation, the gates become a **memory bottleneck**.

Goal

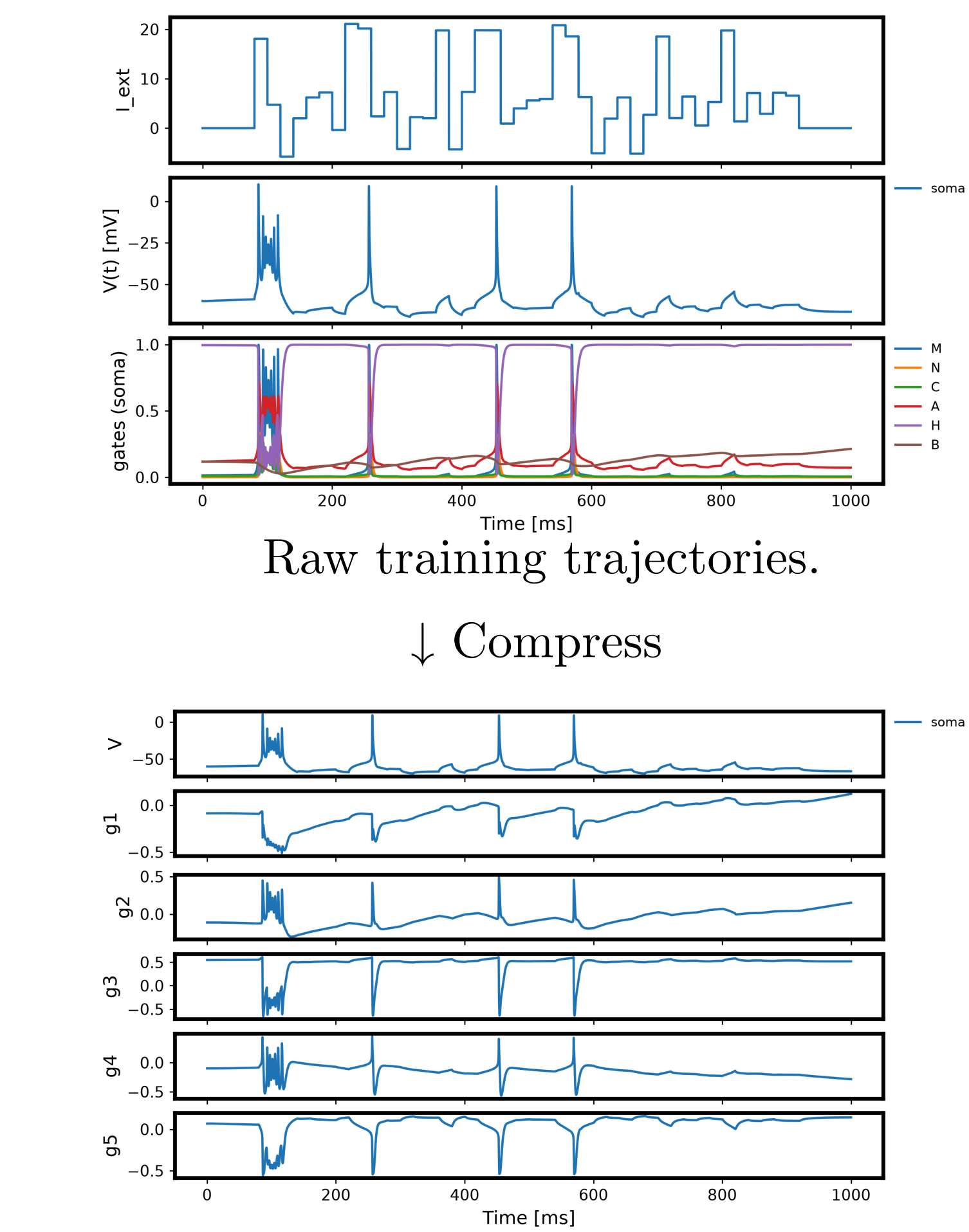
Development of a multi-compartment neuron’s surrogate model capable of reproducing the membrane potential response with few gate variables.

Methods

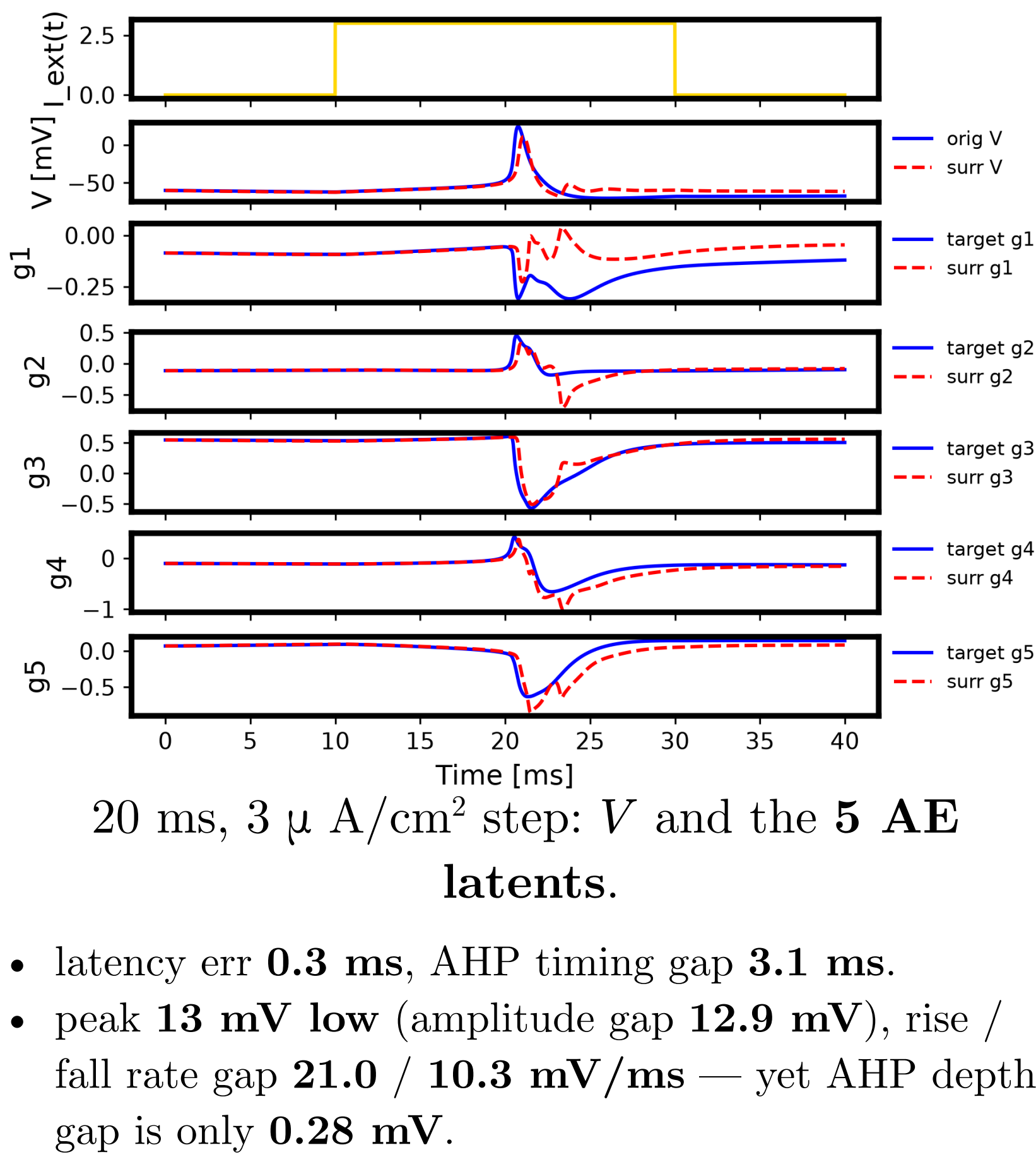


Results and Discussion

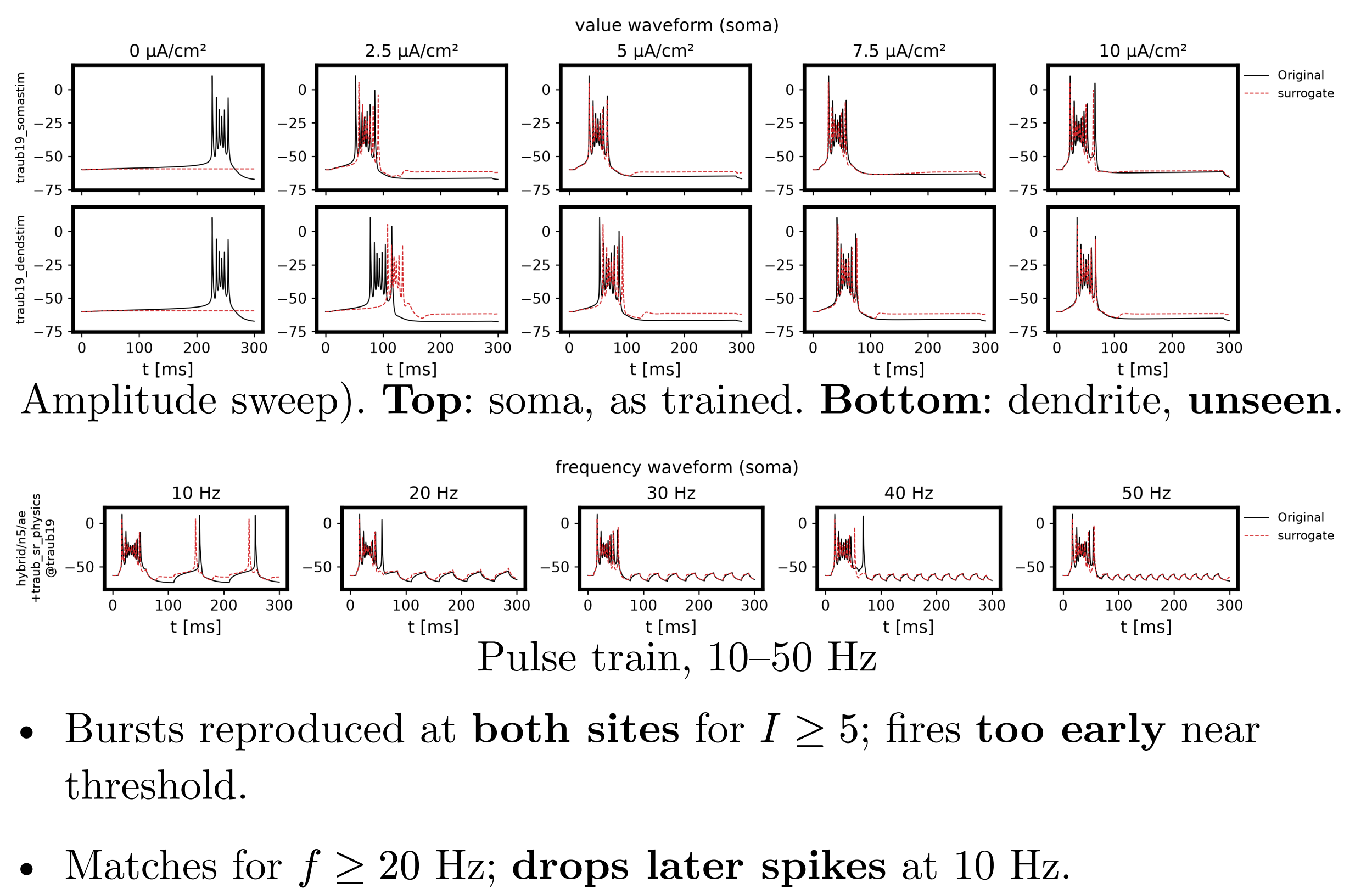
Data



Action Potential reproduction

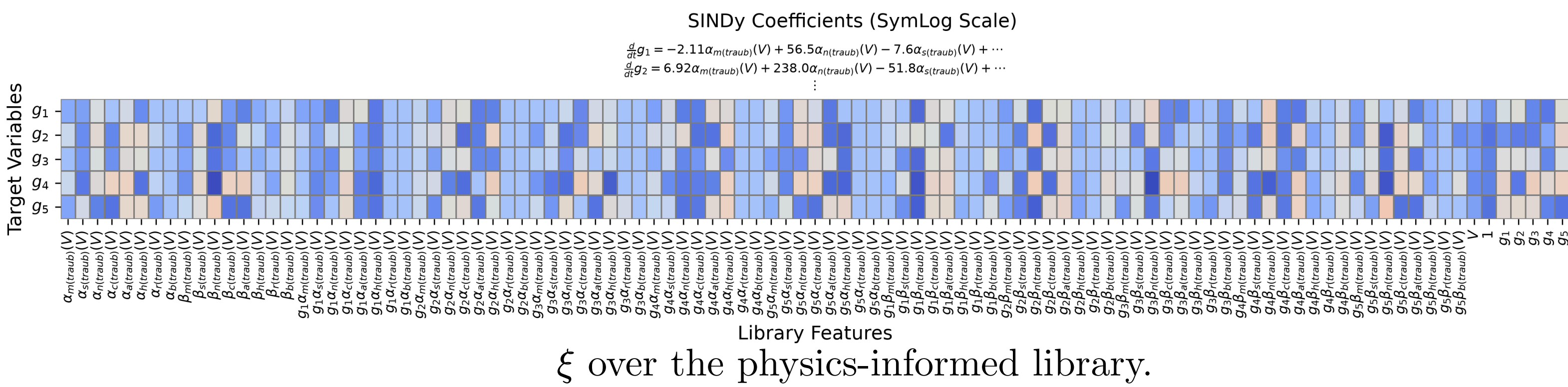


② New stimulus



Latent trajectories used to fit the SINDy library.

④ Identified latent equations



- **79.6% non-zero** — accurate but **not sparse**.

Conclusion

- Latent dynamics preserve **timing**.
 - Error concentrates in the **fast, steep transition** — suggests the **AE encoding (n=5)**, not the SINDy library, fails to preserve it.
 - Replaced **all 19 compartments**, no divergence; transfers to **unseen site • frequency**.
 - **Core finding**: SINDy succeeds **not because we compressed (n: 6→5)** —⑤shows the **same result uncompressed (n = 6)**, so it is the **coordinate change itself** that makes gate dynamics learnable.
 - Cost reduction follows **only if** compression matters — so it remains **unproven**: ξ is still dense (**79.6%** non-zero).
- **Future**: sparsify ξ , push n down, test on other channel models.